

Practical: 4

Selenium Server (Grid) Configuration and Remote Execution

Aim

To install and configure Selenium Grid using Hub-Node architecture and execute Selenium test scripts on multiple browsers using RemoteWebDriver.

Software Requirements

Software	Version (Recommended)
Windows 10/11	64-bit
Java JDK	17 or above
Eclipse IDE	Latest
Selenium Java	4.22 or above
Selenium Server	Latest Stable
Chrome Browser	Latest
Microsoft Edge Browser	Latest
Mozilla Firefox	Latest
ChromeDriver	Matching browser version
GeckoDriver	Latest
EdgeDriver	Matching browser version

Software Download Links

1. Java JDK

<https://www.oracle.com/java/technologies/downloads/>

or

<https://adoptium.net/>

2. Eclipse IDE

<https://www.eclipse.org/downloads/>

3. Selenium Java

<https://www.selenium.dev/downloads/>

4. Selenium Server

Download

selenium-server-4.xx.x.jar

from

<https://github.com/SeleniumHQ/selenium/releases>

5. Chrome Driver

<https://googlechromelabs.github.io/chrome-for-testing/>

6. Firefox Driver (GeckoDriver)

<https://github.com/mozilla/geckodriver/releases>

7. Edge Driver

<https://developer.microsoft.com/en-us/microsoft-edge/tools/webdriver/>

Folder Structure

Create a folder

D:\SeleniumGrid

Inside it

D:\SeleniumGrid

selenium-server.jar

Drivers

chromedriver.exe

geckodriver.exe

msedgedriver.exe

Step 1

Install Java

Open CMD

java -version

Output

java version "17"

If not found

Set JAVA_HOME

Add Java path into Environment Variables.

Step 2

Download Selenium Server

Copy

selenium-server.jar

inside

D:\SeleniumGrid

Step 3

Download Browser Drivers

Copy

chromedriver.exe

geckodriver.exe

msedgedriver.exe

into

D:\SeleniumGrid\Drivers

Step 4

Verify Drivers

CMD

```
chromedriver --version  
geckodriver --version  
msedgedriver --version
```

Step 5

Start Selenium Hub

Open CMD

Navigate

```
cd D:\SeleniumGrid
```

Run

```
java -jar selenium-server.jar hub
```

Output

Started Selenium Grid Hub

Listening on

<http://localhost:4444>

Step 6

Open Grid Console

Browser

<http://localhost:4444>

Grid Dashboard appears.

Step 7

Register Chrome Node

Open another CMD

```
cd D:\SeleniumGrid
```

Run

```
java -jar selenium-server.jar node
```

Since Selenium 4 automatically detects browsers, Chrome will be registered automatically.

Output

Node has been added

Chrome detected

Firefox detected

Edge detected

Step 8

Refresh

<http://localhost:4444>

Now you should see

Chrome

Firefox

Edge

registered.

Step 9

Create Java Project

Open Eclipse

File

New

Java Project

Name

GridExample

Step 10

Add Selenium Libraries

Download Selenium Java ZIP

Extract

Add all JAR files

Also add

libs

folder JARs.

Step 11

Create Package

grid

Step 12

Create Class

RemoteExecution.java

Chrome Test Code

```
package grid;
```

```
import java.net.URL;
```

```
import org.openqa.selenium.WebDriver;
```

```
import org.openqa.selenium.chrome.ChromeOptions;
```

```
import org.openqa.selenium.remote.RemoteWebDriver;
```

```
public class RemoteExecution {
```

```
    public static void main(String[] args) throws Exception {
```

```
        ChromeOptions options = new ChromeOptions();
```

```
        WebDriver driver = new RemoteWebDriver(
```

```
            new URL("http://localhost:4444"),
```

```
            options);
```

```
        driver.get("https://www.google.com");
```

```
        System.out.println(driver.getTitle());
```

```
        driver.quit();
    }
}
```

Firefox Test

```
package grid;

import java.net.URL;

import org.openqa.selenium.WebDriver;
import org.openqa.selenium.firefox.FirefoxOptions;
import org.openqa.selenium.remote.RemoteWebDriver;

public class FirefoxTest {

    public static void main(String[] args) throws Exception {

        FirefoxOptions options = new FirefoxOptions();

        WebDriver driver = new RemoteWebDriver(

            new URL("http://localhost:4444"),

            options);

        driver.get("https://www.google.com");

        System.out.println(driver.getTitle());

        driver.quit();

    }

}
```

Edge Test

```
package grid;

import java.net.URL;

import org.openqa.selenium.WebDriver;
import org.openqa.selenium.edge.EdgeOptions;
import org.openqa.selenium.remote.RemoteWebDriver;

public class EdgeTest {

    public static void main(String[] args) throws Exception {
```

```
        EdgeOptions options = new EdgeOptions();

        WebDriver driver = new RemoteWebDriver(

            new URL("http://localhost:4444"),

            options);

        driver.get("https://www.google.com");

        System.out.println(driver.getTitle());

        driver.quit();

    }

}
```

Step 13

Run

RemoteExecution

Chrome opens remotely.

Output

Google

Run Firefox

Output

Google

Run Edge

Output

Google

Step 14

Observe Grid

Open

http://localhost:4444

You will see

Session Created

Browser Used

Session Finished

Expected Output

Chrome

Google

Firefox

Google

Edge

Google

Cross-Browser Comparison

Feature	Chrome	Firefox	Edge
Browser Launch	Yes	Yes	Yes
Google Opened	Yes	Yes	Yes
Page Title	Google	Google	Google
Test Passed	Yes	Yes	Yes
Speed	Fast	Medium	Fast
Compatibility	Excellent	Good	Excellent

Advantages of Selenium Grid

- Executes tests on multiple browsers.
 - Supports distributed test execution.
 - Reduces execution time through parallel testing.
 - Improves browser compatibility testing.
 - Enables remote execution across different machines.
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Viva Questions

1. What is Selenium Grid?
 2. What is the purpose of a Hub?
 3. What is a Node in Selenium Grid?
 4. What is `RemoteWebDriver`?
 5. Why is Selenium Grid used?
 6. What is cross-browser testing?
 7. What is the default Selenium Grid port?
 8. Can Selenium Grid execute tests in parallel?
 9. Which class is used for remote execution?
 10. What are the advantages of Selenium Grid?
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Result

The Selenium Grid was successfully configured in a Hub–Node architecture. Test scripts were executed remotely using `RemoteWebDriver` on Chrome, Firefox, and Edge browsers. Cross-browser testing was successfully performed, and the application behaved consistently across all tested browsers.

Note: In Selenium Grid 4.x, a single `node` command automatically detects installed browsers (Chrome, Firefox, Edge) and registers them with the Grid. This is the recommended and current approach, even though the architecture is still conceptually referred to as Hub–Node.

To run all browser all together

```
import java.net.URL;
```

```
import org.openqa.selenium.WebDriver;
```

```
import org.openqa.selenium.chrome.ChromeOptions;
import org.openqa.selenium.firefox.FirefoxOptions;
import org.openqa.selenium.remote.RemoteWebDriver;

public class RemoteTest {

    public static void main(String[] args) throws Exception {

        ChromeOptions chrome = new ChromeOptions();

        WebDriver driver = new RemoteWebDriver(
            new URL("http://localhost:4444"),
            chrome);

        driver.get("https://www.google.com");

        System.out.println(driver.getTitle());

        driver.quit();

        FirefoxOptions firefox = new FirefoxOptions();

        driver = new RemoteWebDriver(
            new URL("http://localhost:4444"),
            firefox);

        driver.get("https://www.google.com");

        System.out.println(driver.getTitle());

        driver.quit();
    }
}
```